IRIS FLOWER CLASSIFICATION USING SUPPORT VECTOR MACHINE (SVM)

A Machine Learning Classification Project Report

Prepared by: John Afwilile Kapungu
Project Area: Machine Learning and Data Science
Programming Language: Python
Primary Algorithm: Support Vector Machine (SVM)
Date: October 2026

ABSTRACT

This project investigates the application of supervised machine learning techniques to classify Iris flowers into three species: Iris-setosa, Iris-versicolor, and Iris-virginica. The classification process uses four numerical measurements: sepal length, sepal width, petal length, and petal width. Five machine learning algorithms—K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine (SVM)—were evaluated to identify the best-performing classifier for the experiment.

The dataset contained 150 observations, with 50 observations representing each species. The data were divided into training and testing subsets using an 80:20 stratified split, producing 120 training observations and 30 testing observations. Data exploration, preprocessing, feature analysis, model training, and performance evaluation were conducted using Python and its machine learning ecosystem.

Among the five evaluated algorithms, SVM achieved the highest test accuracy of 96.67%, correctly classifying 29 out of 30 test observations. KNN and Logistic Regression each achieved 93.33%, while Decision Tree and Random Forest each achieved 90.00%. The SVM classification report also recorded a weighted F1-score of approximately 0.97. Based on the available test results, SVM was selected as the final model for this experiment.

The findings demonstrate the potential of supervised machine learning to classify Iris flower species accurately using numerical morphological measurements. However, additional cross-validation and repeated-split evaluation would help establish the consistency and generalizability of the selected model.

Keywords: Machine Learning, Classification, Support Vector Machine, SVM, Iris Dataset, Python, Scikit-learn, Data Science.

CHAPTER ONE: INTRODUCTION

1.1 Background

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or classifications. Classification is a supervised learning task in which a model learns from labelled examples and assigns new observations to predefined categories.

The Iris flower dataset is a widely used dataset in machine learning education and experimentation. It contains measurements of three Iris species and is suitable for studying data preprocessing, exploratory data analysis, model training, and classification evaluation.

This project applies several classification algorithms to the Iris dataset and selects the algorithm with the strongest observed test performance. Support Vector Machine achieved the highest test accuracy among the evaluated models and was therefore selected for detailed discussion.

1.2 Problem Statement

Identifying flower species from physical measurements can be approached as a classification problem. Manual identification may require botanical expertise, while a trained machine learning model can learn relationships between measurements and species labels.

The problem addressed by this project is how to use numerical measurements of Iris flowers to build and evaluate a classification model capable of distinguishing among three species with high predictive accuracy.

1.3 Project Objectives

General objective

To develop and evaluate a machine learning classification model for predicting Iris flower species, selecting the best-performing algorithm from a set of candidate classifiers.

Specific objectives

1. To inspect and understand the Iris dataset and its features.
2. To clean and prepare the data for machine learning.
3. To conduct exploratory data analysis and examine relationships between flower measurements.
4. To train five supervised classification algorithms.
5. To compare the algorithms using test accuracy.
6. To evaluate the selected SVM model using precision, recall, F1-score, and classification results.
7. To recommend improvements for more reliable model evaluation.

1.4 Research Questions

1. Can the four numerical Iris measurements be used to predict flower species accurately?
2. Which of the five evaluated algorithms achieves the highest test accuracy?
3. How effectively does the selected SVM classifier identify each species?
4. What limitations should be considered when interpreting the results?

1.5 Project Scope

The project focuses on classifying the three Iris species using four numerical input features. It covers dataset inspection, exploratory data analysis, preprocessing, model training, comparison, and evaluation. It does not include deployment as a production application or testing on a separate external flower dataset.

CHAPTER TWO: DATASET DESCRIPTION

2.1 Dataset Overview

The Iris dataset used in the experiment contained 150 observations and six original columns.

Attribute	Description
Id	Observation identifier
SepalLengthCm	Sepal length in centimetres
SepalWidthCm	Sepal width in centimetres
PetalLengthCm	Petal length in centimetres
PetalWidthCm	Petal width in centimetres
Species	Target flower species

The four numerical measurements serve as predictor variables, while Species is the target variable.

2.2 Class Distribution

The dataset contains three species with equal representation.

Species	Number of observations	Percentage
Iris-setosa	50	33.33%
Iris-versicolor	50	33.33%
Iris-virginica	50	33.33%
Total	**150**	**100%**

The balanced class distribution supports a straightforward initial comparison of classification performance. However, a balanced dataset does not eliminate the need to evaluate individual class metrics.

2.3 Data Quality

Initial dataset inspection found no missing values and no exact duplicate rows. Exploratory outlier analysis using the interquartile range identified four potential outliers in sepal width.

These observations were not automatically removed because statistical outliers may represent legitimate flower measurements. Removing them without investigating their validity could discard useful information.

2.4 Feature Selection

The Id column was excluded from the predictors because it identifies observations rather than representing a biological measurement. Including it could introduce misleading patterns, particularly if identifiers are ordered by species.

The final predictor matrix was defined as:

- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm

The Species column was used as the target variable.

CHAPTER THREE: METHODOLOGY

3.1 Research Design

The project used a quantitative experimental approach. Multiple supervised classification algorithms were trained and evaluated using the same dataset and test split. Their observed performance was compared to select the final classifier.

3.2 Tools and Technologies

The following Python libraries were used or are appropriate for the documented workflow:

Tool or library	Purpose
Python	Main programming language
Pandas	Data loading and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Preprocessing, model training, and evaluation
Jupyter Notebook	Interactive experimentation

3.3 Data Preprocessing

Preprocessing included inspecting column types, checking for missing values and duplicates, excluding the identifier column, and separating the predictors from the target.

The data were divided using an 80:20 stratified train/test split with a fixed random seed of 42.

Subset	Observations	Percentage
Training data	120	80%
Testing data	30	20%
Total	150	100%

Stratification preserves the approximate class proportions in both subsets. Fixing the random seed supports reproducibility.

3.4 Feature Scaling

SVM is sensitive to the scale of input features. Standardization transforms numerical features so that each feature is represented on a comparable scale.

The StandardScaler was used in a pipeline with SVM. This approach ensures that the scaler is fitted on the training data during model fitting rather than using information from the test set.

The same principle should be applied to KNN and Logistic Regression when using standardization. Decision Tree and Random Forest generally do not require feature scaling.

3.5 Exploratory Data Analysis

Exploratory data analysis was used to understand feature distributions and relationships before modelling. Appropriate visualizations included:

- Histograms and distribution plots to examine numerical measurements.
- Boxplots to inspect potential outliers.
- Pairplots to explore relationships between features and species.
- Correlation heatmaps to examine linear relationships among numerical variables.

The exploratory analysis indicated a strong positive correlation between petal length and petal width, approximately 0.963. These features are potentially useful for species discrimination, although correlation alone does not establish causal relationships or determine feature importance.

3.6 Classification Algorithms

Five supervised learning algorithms were evaluated.

K-Nearest Neighbors (KNN): Classifies an observation using the labels of nearby training observations. Its performance depends on the choice of neighbour count and the distance metric. Feature scaling is important.

Decision Tree: Learns a sequence of feature-based decisions. It is relatively easy to interpret, but its performance can depend on tree depth and other regularization settings.

Logistic Regression: Estimates class probabilities using a linear decision function. It provides a useful baseline and can perform well when class boundaries are sufficiently separable.

Random Forest: Combines predictions from multiple decision trees. It can model nonlinear relationships and reduce the variance of a single tree, although performance depends on its configuration.

Support Vector Machine (SVM): Learns decision boundaries that separate classes. The radial basis function (RBF) kernel allows nonlinear boundaries and was used for the selected SVM model.

3.7 Model Selection Procedure

The five algorithms were trained and compared using the same held-out test set. Test accuracy was used as the primary comparison metric.

The observed results identified SVM as the highest-accuracy model. The detailed SVM classification report was then used to examine performance by species.

For a more robust assessment, cross-validation should also be used for all candidate models. If hyperparameter tuning is performed, it should use training data only, leaving the test set untouched until the final evaluation.

CHAPTER FOUR: RESULTS AND ANALYSIS

4.1 Comparative Model Performance

The following test accuracies were obtained from the experiment.

Rank	Algorithm	Test accuracy	Correct predictions
1	**Support Vector Machine (SVM)**	**96.67%**	29/30
2	K-Nearest Neighbors (KNN)	93.33%	28/30
2	Logistic Regression	93.33%	28/30
4	Decision Tree	90.00%	27/30
4	Random Forest	90.00%	27/30

SVM achieved the highest observed test accuracy. Its score was approximately 3.34 percentage points higher than KNN and Logistic Regression, and 6.67 percentage points higher than Decision Tree and Random Forest, based on the reported rounded accuracies.

These differences correspond to one additional correct prediction compared with KNN and Logistic Regression, and two additional correct predictions compared with Decision Tree and Random Forest.

4.2 Selected Model: Support Vector Machine

SVM was selected because it achieved the highest test accuracy among the five evaluated algorithms. The reported classification results were:

Metric	Score
Test accuracy	96.67%
Macro precision	0.97
Macro recall	0.97
Macro F1-score	0.97
Weighted precision	0.97
Weighted recall	0.97
Weighted F1-score	0.97

The classification report values are rounded to two decimal places. The test accuracy corresponds to 29 correct predictions out of 30 observations.

4.3 Class-Level Evaluation

Species	Precision	Recall	F1-score	Support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	0.90	0.95	10
Iris-virginica	0.91	1.00	0.95	10

Iris-setosa: The model correctly classified all 10 test observations. Precision, recall, and F1-score were all 1.00.

Iris-versicolor: Precision was 1.00, meaning every observation predicted as Versicolor was correct. Recall was 0.90, indicating that 9 of the 10 actual Versicolor observations were identified correctly.

Iris-virginica: Recall was 1.00, indicating that all 10 actual Virginica observations were identified. Precision was 0.91, suggesting that one observation from another class was incorrectly predicted as Virginica.

The results show excellent performance on the test observations, with the single error involving confusion between species rather than a failure to identify every class.

4.4 Interpretation of Findings

The selected SVM classifier performed well on the four numerical flower measurements. Its high test accuracy suggests that these measurements contain useful information for distinguishing the three species.

The strong results are consistent with the suitability of SVM for classification tasks where classes can be separated using an appropriate decision boundary. However, the experiment alone does not prove that SVM will outperform every alternative on other train/test splits or unseen datasets.

The test set contains only 30 observations, so one prediction represents approximately 3.33 percentage points of accuracy. The results should therefore be interpreted as evidence of strong performance on this particular split, not a guarantee of identical performance in future applications.

CHAPTER FIVE: DISCUSSION

5.1 Why SVM Was Selected

SVM was selected because it achieved the highest test accuracy among the five evaluated algorithms. It correctly classified 29 of 30 test observations and achieved a weighted F1-score of approximately 0.97.

The RBF kernel can represent nonlinear decision boundaries, which can be useful when the relationships between numerical measurements and species labels are not adequately represented by a simple linear boundary.

The selection is based on observed predictive performance, not on a claim that SVM is universally superior to other classification algorithms.

5.2 Comparison with Alternative Algorithms

KNN and Logistic Regression achieved 93.33% test accuracy, making them competitive alternatives. KNN offers a conceptually simple distance-based approach, while Logistic Regression provides a comparatively interpretable linear classification model.

Decision Tree and Random Forest each achieved 90.00% test accuracy under the configurations used in this experiment. Their lower scores on this split do not imply that these algorithms are inherently unsuitable for the Iris dataset; different configurations or data splits may produce different results.

5.3 Implications for Machine Learning Practice

The project illustrates several important principles:

1. Data quality and appropriate feature selection should be addressed before training.
2. An identifier should not be treated as a meaningful predictor without justification.
3. Scaling should be incorporated into the training pipeline for scale-sensitive algorithms.
4. Multiple algorithms should be evaluated using consistent data splits and metrics.
5. Model selection should consider cross-validation, not test accuracy alone.
6. Per-class metrics provide information that overall accuracy cannot show by itself.

CHAPTER SIX: LIMITATIONS

6.1 Small Test Set

Only 30 observations were used for testing. This makes the measured accuracy sensitive to a small number of predictions and limits the precision of the performance estimate.

6.2 Limited Evaluation Splits

The reported comparison uses one train/test split. The ranking may change with a different split. Repeated stratified cross-validation would provide a more reliable estimate of average performance and variability.

6.3 Hyperparameter Dependence

Algorithm performance depends on model configurations, including the SVM kernel and its C and gamma parameters. The available results do not include the selected hyperparameter values or mean cross-validation score, so these should not be reported as confirmed findings.

6.4 Dataset Generalizability

The experiment used a small, well-known dataset with three species and four numerical features. Performance on this dataset does not establish performance on other flower species, different measurement conditions, or larger real-world datasets.

6.5 Limited Evidence for Statistical Superiority

The difference between SVM and KNN or Logistic Regression is one correct prediction on the test set. This difference is too small, on its own, to establish that SVM is statistically or consistently superior.

CHAPTER SEVEN: RECOMMENDATIONS AND FUTURE WORK

7.1 Recommendations

1. **Retain SVM as the current selected model.** It achieved the highest observed test accuracy of 96.67%.
2. **Validate using cross-validation.** Compare the mean and standard deviation of stratified cross-validation scores for all five algorithms.
3. **Tune hyperparameters carefully.** Use training data and cross-validation to tune SVM parameters without using the test set for model selection.
4. **Inspect the confusion matrix.** Use it to identify exactly which species were confused and to support the class-level interpretation.
5. **Preserve a reproducible workflow.** Record the random seed, feature columns, preprocessing steps, model configuration, and evaluation metrics.
6. **Save the trained pipeline.** If the model will be reused, save the fitted preprocessing and classifier together so that new observations are processed consistently.
7. **Document limitations.** Present the test score as a result from this experiment rather than a guarantee of future accuracy.

7.2 Future Work

Future improvements could include:

- Applying stratified cross-validation to all candidate models.
- Comparing repeated train/test splits.
- Recording the best SVM hyperparameters and its mean cross-validation score.
- Evaluating additional metrics, including per-class precision, recall, and F1-score.
- Creating a simple web application that accepts flower measurements and predicts the species.
- Testing the saved model with valid, unseen flower measurements.
- Publishing the code, dataset source, dependencies, and results in a version-controlled repository.

CHAPTER EIGHT: CONCLUSION

This project developed and evaluated a supervised machine learning workflow for classifying Iris flowers using four numerical measurements. Five algorithms were compared: KNN, Decision Tree, Logistic Regression, Random Forest, and SVM.

Among the evaluated algorithms, SVM achieved the highest observed test accuracy of 96.67%, correctly classifying 29 of the 30 test observations. KNN and Logistic Regression each achieved 93.33%, while Decision Tree and Random Forest each achieved 90.00%. The SVM classification report showed a weighted F1-score of approximately 0.97, with perfect classification metrics for Iris-setosa.

Based on the available test results, SVM is the selected model for this experiment. Nevertheless, the small test set and the use of a single split mean that further validation is needed before concluding that its advantage is consistent. Cross-validation, careful hyperparameter tuning, and repeated evaluation would strengthen confidence in the final model.

Overall, the project demonstrates how Python and Scikit-learn can be used to perform data exploration, preprocessing, supervised learning, model comparison, and classification evaluation in a reproducible machine learning workflow.

REFERENCES

1. Fisher, R. A. (1936). *The use of multiple measurements in taxonomic problems*. Annals of Eugenics, 7(2), 179–188.
2. Scikit-learn Developers. *Scikit-learn: Machine Learning in Python*. https://scikit-learn.org/
3. Scikit-learn Developers. *Support Vector Machines*. https://scikit-learn.org/stable/modules/svm.html
4. Scikit-learn Developers. *Model Evaluation: Quantifying the Quality of Predictions*. https://scikit-learn.org/stable/modules/model_evaluation.html
5. UCI Machine Learning Repository. *Iris Dataset*. https://archive.ics.uci.edu/dataset/53/iris

Prepared by: John Afwilile Kapungu

Project: Iris Flower Classification Using Machine Learning
Selected Algorithm: Support Vector Machine (SVM)
Project Status: Model evaluation completed; SVM selected with 96.67% test